University of Warwick

Module Code: IB94X0 Business Statistics
School Year: 2026/27

Individual Assignment 1 Question and Guidance

The Structure

The assignment should be completed in RMarkdown and knitted into a html file. Your submission should be a zip file containing the RMarkdown file and the html output it creates. When the RMarkdown file is re-run, it must produce the same html output that you submit.

This should have two parts, as indicated in *template_assignment1.rmd.*

The first part should include the code that you produce to perform all stages of the data analysis. It should read in the datafile provided, check the integrity of data, calculate all of the statistics requested, and produce all of the figures/visualisations requested. The purpose of this first part is that it could be shared with someone else who is not familiar with the data or project, but is expert in R, tidyverse, and statistics. They should be able to see how you achieved your results, and if necessary, easily modify the code to work with new or updated data. Therefore, the appropriateness of your comments and data documentation will be assessed, as well as clarity and consistency of approach such as consistent use of compatible packages.

The second part should be a short, polished and professional report presenting and interpreting the findings for an audience of managers or policy makers. You should assume that the managers/policy makers have a basic knowledge of the data and where it comes from, but are not experts in statistics or R. Therefore, in addition to reporting the results of statistical tests using standard reporting conventions, you should also state the results in plain English and say what they mean or what they tell us about the real scenario – in this case what they tell us about the temperature on the different London Underground Lines. The visualisations and figures in this section should be as clear as possible and of publication standard.

The official word count for this assignment is 1500 words, but the written elements of the report **should be much shorter than this**. Since this is a code and analysis-based assignment, the word count is not directly applicable in the way that it is for an essay. The word count is provided as a guide: that this assignment is assumed to be approximately the same amount of work as a 1500-word essay. The comments (in code and between code blocks) in the first part should be short, brief, and sufficient for another data analyst to quickly determine what you are doing and why. The conclusions and writing in your second part should be no more than **300 words**. If you include more than this then you are likely including irrelevant information. Including irrelevant information or irrelevant code will have a negative impact upon your grade.

The details of what you should do are provided on the next page.

The Request

London Underground Temperatures

We are taking the role of a policy maker in London who is running a pilot scheme and is deciding where to install a new air-cooling system on the London Underground Lines. To do this we are going to use monthly average temperatures from the London Underground Lines:

Load in the data and check it for integrity/quality and any errors. If you elect to remove any observations, explain why.

Provide a summary table that shows the following, both across the entire dataset, and for each individual London Underground line, where temperature levels were measured:
- The mean temperature
- The median temperature
- The standard deviation of the temperature
- The lowest temperature recorded for that underground Line
- The highest temperature recorded for that underground Line

The data set includes older London Underground Lines, these Lines are unusual in that they are built closer to the surface than the other lines – these are grouped together and called "Sub-Surface Lines". You should exclude these from the subsequent analysis.

Plot the distribution of temperature across one particular year in the dataset (you are free to choose that year).

This should show separate distributions for each line, as well as a plot pooling across all lines.

Perform a t-test to compare the temperature on the Jubilee Line to the Central Line.

Perform an additional t-test to compare the temperature on the Jubilee Line to the Central Line only in the summer months (i.e. May, June, July and August).

What would you advise the planner about where they should trail a new air-cooling system; the Jubilee Line or the Central Line?

Marking Criteria

Comprehension
Demonstrate understanding of the data and the purpose of the analysis steps taken. Appropriately interpret both the task and the implications of the results.

Analysis & Critical Evaluation
Apply appropriate checks and validation of the data and apply appropriate analyses. Where appropriate, critique and evaluate the data and results.

Academic Writing

Clear description of the analysis and results that is appropriate for the stated audience(s). Concise description that clearly identifies and communicates the important and relevant elements. High quality visualisations that demonstrate the desired information or results.

Technical Capability

Clear and appropriate data analysis with well structured code and appropriate commenting.

Maximum Length:	**1500 words** *(see main text above)*
Assessment Weighting:	**20%**
Submission Deadline:	**29 October 2026 12:00 (Midday, UK time)**
Artificial Intelligence:	*PERMITTED*
